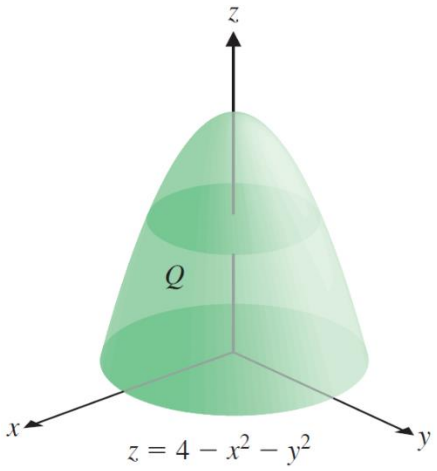
Vector Calculus and Complex Analysis Homework 5

1. Consider the solid Q bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy-plane. Verify divergence theorem for $\boldsymbol{F}(x, y, z) = \langle x^3, y^3, z^3 \rangle$.



Solution

The parameterization of the curved surface of Q is $\boldsymbol{r}(u,v) = (u, v, 4 - u^2 - v^2)$.

$$\boldsymbol{r}_u = (1, 0, -2u), \quad \boldsymbol{r}_v = (0, 1, -2v)$$

The outward normal vector is $\boldsymbol{r}_u \times \boldsymbol{r}_v = (1, 0, -2u) \times (0, 1, -2v) = (2u, 2v, 1)$

$$\begin{aligned}
\iint_{\substack{\text{curved} \\ \text{surface}}} \langle x^3, y^3, z^3 \rangle \cdot N dA &= \iint_{u^2+v^2 \le 4} \langle u^3, v^3, (4 - u^2 - v^2)^3 \rangle \cdot \langle 2u, 2v, 1 \rangle \, du dv \\
&= \iint_{u^2+v^2 \le 4} \langle u^3, v^3, (4 - u^2 - v^2)^3 \rangle \cdot \langle 2u, 2v, 1 \rangle \, du dv \\
&= \int_0^{2\pi} \int_0^2 \left(2r^4 \cos^4\theta + 2r^4 \sin^4\theta + (4 - r^2)^3 \right) r \, dr \, d\theta \\
&= \int_0^{2\pi} \left[\tfrac{2}{6} r^6 (\cos^4\theta + \sin^4\theta) - \tfrac{1}{8} (4 - r^2)^4 \right]_0^2 d\theta \\
&= \int_0^{2\pi} \left[\tfrac{64}{3} (\cos^4\theta + \sin^4\theta) + 32 \right] d\theta \\
&= \int_0^{2\pi} \left[\tfrac{64}{3} \left(\left(\tfrac{1+\cos 2\theta}{2} \right)^2 + \left(\tfrac{1-\cos 2\theta}{2} \right)^2 \right) + 32 \right] d\theta \\
&= \int_0^{2\pi} \left[\tfrac{64}{3} \left(\tfrac{1+\cos^2 2\theta}{2} \right) + 32 \right] d\theta \\
&= \int_0^{2\pi} \left[\tfrac{64}{3} \left(\tfrac{1+0.5(1+\cos 4\theta)}{2} \right) + 32 \right] d\theta \\
&= \left(\tfrac{64}{3} \cdot \tfrac{1.5}{2} + 32 \right) 2\pi \\
&= 96\pi
\end{aligned}$$

The parameterization of the base of Q is $\boldsymbol{r}(x,y)=(x,y,0), \quad x^2+y^2\le 4.$

$$\boldsymbol{r}_x=(1,0,0), \quad \boldsymbol{r}_y=(0,1,0)$$

The outward normal vector is $\boldsymbol{r}_y\times\boldsymbol{r}_x=(0,1,0)\times(1,0,0)=(0,0,-1)$

$$\iint_{\text{base}}\left\langle x^3,y^3,z^3\right\rangle\cdot N\,dA=\iint_{\text{base}}\left\langle x^3,y^3,0\right\rangle\cdot(0,0,-1)\,dA=0$$

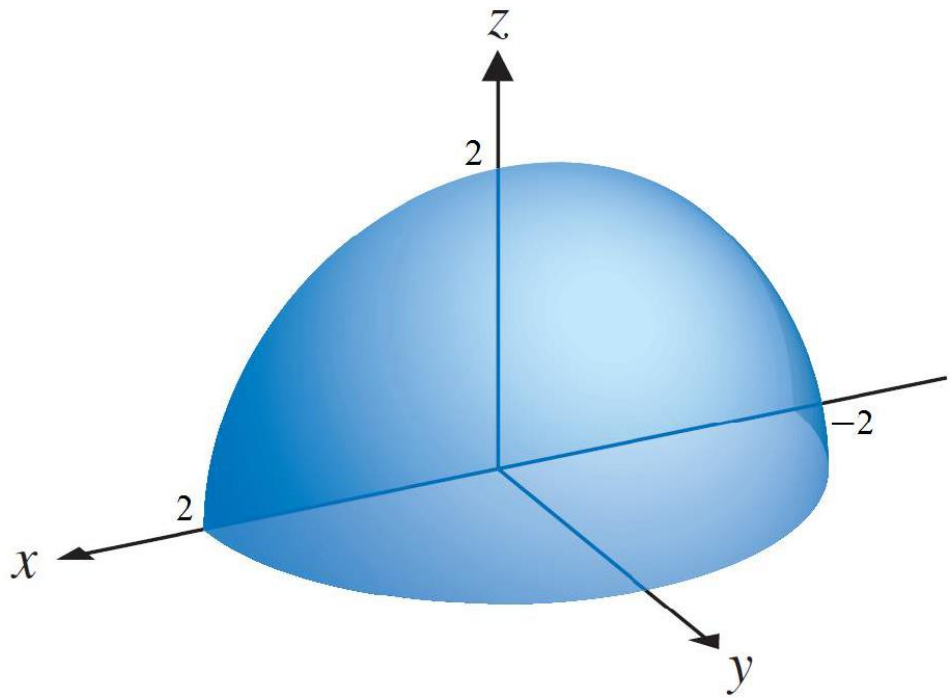
$$\iint_{\partial Q}\left\langle x^3,y^3,z^3\right\rangle\cdot N\,dA=\iint_{\substack{\text{curved}\\ \text{surface}}}\left\langle x^3,y^3,z^3\right\rangle\cdot N\,dA+\iint_{\text{base}}\left\langle x^3,y^3,z^3\right\rangle\cdot N\,dA=96\pi$$

$$\nabla\cdot\mathbf{F}(x,y,z)=\nabla\cdot\langle x^3,y^3,z^3\rangle=3x^2+3y^2+3z^2.$$

From the Divergence Theorem,

$$\begin{aligned}\iint_{\partial Q}\mathbf{F}\cdot\mathbf{n}\,dS&=\iiint_{Q}\nabla\cdot\mathbf{F}(x,y,z)\,dV\\&=\iiint_{Q}(3x^2+3y^2+3z^2)\,dV\\&=3\int_0^{2\pi}\int_0^{2}\int_0^{4-r^2}(r^2+z^2)\,r\,dz\,dr\,d\theta\\&=3\int_0^{2\pi}\int_0^{2}\left(r^2z+\frac{z^3}{3}\right)\Big|_{z=0}^{z=4-r^2}r\,dr\,d\theta\\&=3\int_0^{2\pi}\int_0^{2}\left[r^3(4-r^2)+\frac{1}{3}(4-r^2)^3r\right]dr\,d\theta\\&=96\pi\end{aligned}$$

2. Consider the solid Q satisfied $y, z \geq 0, x^2 + y^2 + z^2 \leq 4$. Verify divergence theorem for $\boldsymbol{F}(x, y, z) = \langle x^3, y^3, z^3 \rangle$.



Solution

The parameterization of the curved surface of Q is

$$\boldsymbol{r}(u, v) = (2\cos v \cos u, 2\cos v \sin u, 2\sin v), \quad 0 \leq u \leq \pi, 0 \leq v \leq \tfrac{\pi}{2}$$

The outward normal vector is $\boldsymbol{r}_u \times \boldsymbol{r}_v = (4\cos^2 v \cos u, 4\cos^2 v \sin u, 4\cos v \sin v)$

$$\iint_{\substack{\text{curved}\\ \text{surface}}} \left\langle x^3, y^3, z^3 \right\rangle \cdot N dA$$

$$= \iint_{\substack{0\le u\le \pi,\\ 0\le v\le \frac{\pi}{2}}} \left\langle (2\cos v\cos u)^3, (2\cos v\sin u)^3, (2\sin v)^3 \right\rangle \cdot \left\langle 4\cos^2 v\cos u, 4\cos^2 v\sin u, 4\cos v\sin v \right\rangle du dv$$

$$= 32 \iint_{\substack{0\le u\le \pi,\\ 0\le v\le \frac{\pi}{2}}} \cos^5 v\cos^4 u + \cos^5 v\sin^4 u + \cos v\sin^4 v \, du dv$$

$$= 32\int_0^{\pi}\left[\int_0^{\pi/2}\left((1-\sin^2 v)^2(\cos^4 u + \sin^4 u) + \sin^4 v\right) d\sin v\right] du$$

$$= 32\int_0^{\pi}\left[(\cos^4 u + \sin^4 u)\left(\sin v - 2\frac{\sin^3 v}{3} + \frac{\sin^5 v}{5}\right) + \frac{\sin^5 v}{5}\right]_0^{\pi/2} du$$

$$= 32\int_0^{\pi}\left[(\cos^4 u + \sin^4 u)\left(\frac{8}{15}\right) + \frac{1}{5}\right] du$$

$$= 32\int_0^{\pi}\left[\tfrac{8}{15}\left(\tfrac{1+0.5(1+\cos 4\theta)}{2}\right) + \tfrac{1}{5}\right] du$$

$$= \frac{96}{5}\pi$$

$$\iint_{\text{back}} \left\langle x^3, y^3, z^3 \right\rangle \cdot N dA = \iint_{\text{base}} \left\langle x^3, y^3, z^3 \right\rangle \cdot N dA = 0 \quad \Rightarrow \quad \iint_{\partial Q} \left\langle x^3, y^3, z^3 \right\rangle \cdot N dA = \frac{96}{5}\pi$$

$$\iiint_Q \nabla \boldsymbol{F}\, dV = \iiint_Q 3(x^2 + y^2 + z^2) dV$$

$$= 3\int_{-2}^{2}\int_0^{\sqrt{4-x^2}}\int_0^{\sqrt{4-x^2-y^2}} (x^2 + y^2 + z^2) dz dy dx$$

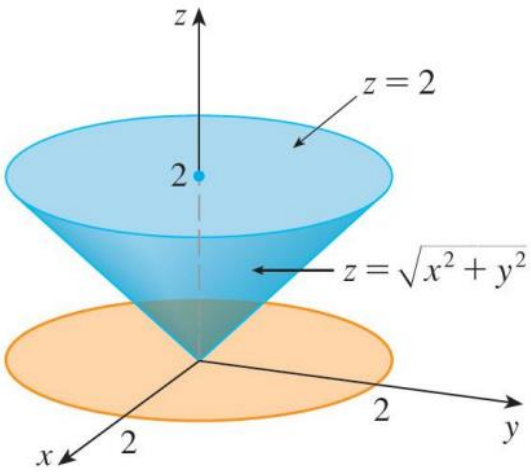
$$= 3\int_0^{\pi}\int_0^{\pi/2}\int_0^{2} \rho^2(\rho^2\sin\varphi) d\rho\, d\varphi\, d\theta$$

$$= 3\int_0^2 \rho^4 d\rho \int_0^{\pi/2} \sin\varphi d\varphi \int_0^{\pi} 1 d\theta$$

$$= 3\left[\frac{\rho^5}{5}\right]_0^2 \cdot \left[-\cos\varphi\right]_0^{\pi/2} \cdot \pi$$

$$= \frac{96}{5}\pi$$

3. Consider the cone Q in the below figure. Verify divergence theorem for $\boldsymbol{F}(x,y,z)=\langle xy^2, x^2y, 0\rangle$.



Solution

The parameterization of the curved surface of Q is

$$\boldsymbol{r}(u,v)=(u\cos v, u\sin v, u), \quad 0\le u\le 2, 0\le v\le 2\pi$$

The outward normal vector is $-\boldsymbol{r}_u\times\boldsymbol{r}_v=-(\cos v,\sin v,1)\times(-u\sin v, u\cos v, 0)=(u\cos v, u\sin v, -u)$

$$\iint_{\substack{\text{curved}\\ \text{surface}}} \langle xy^2, x^2y, 0\rangle\cdot N dA$$

$$=\iint_{\substack{0\le u\le 2,\\ 0\le v\le 2\pi}} \langle u^3\cos v\sin^2 v, u^3\cos^2 v\sin v, 0\rangle\cdot\langle u\cos v, u\sin v, -u\rangle du dv$$

$$=\int_0^{2\pi}\int_0^2 2u^4\cos^2 v\sin^2 v du dv$$

$$=\frac{64}{5}\int_0^{2\pi}\cos^2 v\sin^2 v dv$$

$$=\frac{64}{5}\int_0^{2\pi}\left(\frac{1+\cos 2v}{2}\right)\left(\frac{1-\cos 2v}{2}\right)dv$$

$$=\frac{16}{5}\int_0^{2\pi}(1-\tfrac{1+\cos 4v}{2})dv$$

$$=\frac{16}{5}\cdot\frac{1}{2}\cdot 2\pi=\frac{16}{5}\pi$$

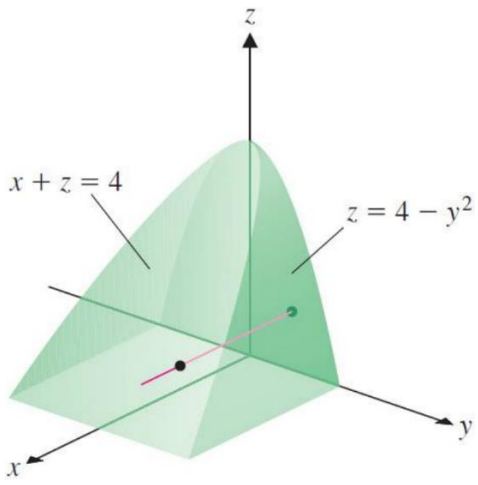
The outward normal vector of disk of Q is (0,0,1)

$$\iint_{\text{disk}} \langle xy^2, x^2y, 0\rangle \cdot N dA = 0$$

$$\iint_{\partial Q} \langle xy^2, x^2y, 0\rangle \cdot N dA = \frac{16}{5}\pi$$

$$\begin{aligned}
\iiint_Q \nabla \boldsymbol{F}\, dV &= \iiint_Q (x^2+y^2) dV \\
&= \int_0^{2\pi}\int_0^2\int_r^2 r^2 r\, dz\, dr\, d\theta \\
&= \int_0^2 r^3(2-r)dr \int_0^{2\pi} 1 d\theta \\
&= 2\pi\left[\tfrac{1}{2}r^4 - \tfrac{1}{5}r^5\right]_0^2 \\
&= \frac{16}{5}\pi
\end{aligned}$$

4. Consider the solid Q bounded by $z = 4-y^2, x+z=4, x=0$ and $z = 0$. Verify divergence theorem for $\boldsymbol{F}(x,y,z) = \langle 0, xy, yz\rangle$.



Solution

$$Q = \{(x,y,z) \mid 0 \le x \le 4-z, 0 \le z \le 4-y^2, -2 \le y \le 2\}$$

The outward normal vector of base of Q is $(0,0,-1)$.

$$\iint_{\text{base}} \langle 0, xy, yz\rangle \cdot N dA = \iint_{\text{base}} -yz dA = 0$$

The outward normal vector of back of Q is $(-1,0,0)$.

$$\iint_{\text{back}} \langle 0, xy, yz\rangle \cdot N dA = \iint_{\text{back}} 0 dA = 0$$

The outward normal vector of plane $x + z = 4$ of Q is (1,0,1).

$$\iint_{x+z=4} \langle 0, xy, yz\rangle \cdot N dA = \iint_{x+z=4} yz dA = \int_{-2}^{2}\int_{0}^{4-y^2} yz dz dy = \frac{1}{2}\int_{-2}^{2} y\left[z^2\right]_0^{4-y^2} dy = \frac{1}{2}\int_{-2}^{2} y(4-y^2)^2 dy = 0$$

The parameterization of the curved surface of Q is

$$\boldsymbol{r}(u,v) = (u, v, 4-v^2), \quad 0 \le u \le 4-(4-v^2) = v^2, -2 \le v \le 2\}$$

The outward normal vector is $\boldsymbol{r}_u \times \boldsymbol{r}_v = (1,0,0)\times(0,1,-2v) = (0,2v,1)$

$$\iint_{\substack{\text{curved}\\ \text{surface}}} \langle 0, xy, yz\rangle \cdot N dA = \int_{-2}^{2}\int_{0}^{v^2} \langle 0, uv, v(4-v^2)\rangle \cdot \langle 0, 2v, 1\rangle du dv$$

$$= \int_{-2}^{2}\int_{0}^{v^2} \left(2uv^2 + v(4-v^2)\right) du dv$$

$$= \int_{-2}^{2}\left[u^2v^2 + uv(4-v^2)\right]_0^{v^2} dv$$

$$= \int_{-2}^{2}\left[v^6 + v^3(4-v^2)\right] dv$$

$$= \left[\tfrac{1}{7}v^7 + \tfrac{1}{4}v^4 - \tfrac{1}{6}v^6\right]_{-2}^{2}$$

$$= \tfrac{256}{7}$$

Hence $\iint_{\partial Q} \langle 0, xy, yz\rangle \cdot N dA = \tfrac{256}{7}$

$$\iiint_Q \operatorname{div}(\boldsymbol{F}) dV = \iiint_Q (x+y) dV$$

$$= \int_{-2}^{2}\int_{0}^{4-y^2}\int_{0}^{4-z} (x+y) dx dz dy$$

$$= \int_{-2}^{2}\int_{0}^{4-y^2} \left(\tfrac{1}{2}(4-z)^2 + y(4-z)\right) dz dy$$

$$= -\int_{-2}^{2}\left[\tfrac{1}{6}(4-z)^3 + \tfrac{1}{2}y(4-z)^2\right]_0^{4-y^2} dy$$

$$= -\int_{-2}^{2}\left[\tfrac{1}{6}y^6 + \tfrac{1}{2}y^5 - \tfrac{1}{6}4^3 - \tfrac{1}{2}4^2 y\right] dy$$

$$= -\left[\tfrac{1}{42}y^7 + \tfrac{1}{12}y^6 - \tfrac{1}{6}4^3 y - \tfrac{1}{4}4^2 y^2\right]_{-2}^{2}$$

$$= -\left[\tfrac{256}{42} - \tfrac{1}{6}4^4\right]$$

$$= \tfrac{256}{7}$$